

# Sinusoid

Montague Hamilton

1 October 2026

## 1 Questions

1.

$$h(t) = c + a \sin\left(\frac{2\pi t}{p} + \phi\right)$$

For a wheel centred at height 4, radius 3, period 12; the rider starts at the top. At  $t = 3$ ,

$$\begin{aligned} h(t) &= 4 + 3 \sin\left(\frac{2\pi 3}{12} + \frac{\pi}{2}\right) \\ &= 4 + 3 \sin(\pi) \\ &= 4 \end{aligned}$$

So at the same level as the wheels centre.